

Steps

The resources for this week include lists of options to utilize for your list. The list is not all of the options. You can utilize this list or use other software of your choice.

1: Start a Word document, making sure you start with your name, the assignment name and the due date.

2: In the first paragraph, define the company for which you will be setting up the servers. Include the company's industry and any product or services in the description. You should also state the number of employees that are remote and the number that are in the main office.

- This company can be fictitious or can be a company with which you are familiar.

3: Evaluate the options for email and web servers. For each server type, choose an operating system (Linux or Windows) and software to be used.

4: In the document, write a section that is your choice for the email server.

- Add a paragraph, discussing your justification for your choice of operating system and email server software.
- Define the hardware requirements for the server to effectively run the operating system and software.

5: Document a second section that is your choice for the web server.

- Add a paragraph, discussing your justification for your choice of operating system and web server software.
- Define the hardware requirements for the server to effectively run the operating system and software.

6: In your conclusion paragraph, discuss how the company's industry and products affected your decision. Also discuss what would change if the number of employees doubled over the next year.

Homey Analytics

My made-up company will be called Homey Analytics, a small data analytics and consulting firm. My company will provide data visualization techniques, modeling services, and cloud-based reporting to retail clients. We will rely heavily on communication with our clients and a professional web presence to show off our services to potential clients.

The company currently has 40 employees, with 25 employees working remotely and 15 employees located in the main office. We are based in New York city as there are many potential clients there. Since many employees work remotely and interact with clients on a day-to-day basis, secure email communication and reliable web hosting are extremely important.

My thought process:

Potential Email server options:

- Linux
 - Commonly used for email servers
 - Stable
 - Secure
 - cost-effective
- Windows Server
 - Goes well with Microsoft-based infrastructure
 - Pairs with Active Directory

Potential Email Software Options:

- Microsoft Exchange Server
 - Works with Windows tools
- Postfix
 - SMTP
 - MX
- Dovecot
 - Works well with small installations
 - IMAP

Microsoft Exchange integrates well with Microsoft 365 tools and Active Directory but comes with higher licensing costs.

Postfix and Dovecot offer flexibility and lower cost but require more manual configuration and maintenance.

Efficiency and Pricey VS Timely and Flexibility

Potential Operating System Options:

- Linux probably Ubuntu Server
- Windows Server

Potential Web Server Software Options:

- Apache HTTP Server
 - o Works with both Linux and Windows
 - o Flexible
- Nginx
 - o Works with both Linux and Windows
 - o Great performance
- Microsoft IIS
 - o I've used this before for classes and internships so I'm familiar
 - o Solid UI

Linux web servers using Apache or Nginx are widely used, secure, and cost-effective.

Windows Server with IIS integrates pairs with ASP.NET applications and Microsoft environments but also have high licensing costs.

Companies hosting modern web applications usually go with Linux with Nginx or Apache.

Chosen configuration:

Email Server Implementation:

Operating System:

- Linux (Ubuntu Server LTS)

Email Software

- Postfix (SMTP and MX)
- Dovecot (IMAP)

Homey Analytics does not need to rely heavily on Microsoft-specific services, so a Linux-based solution gives us cost savings and flexibility. Ubuntu Server LTS has long-term support and strong security. Postfix and Dovecot are already used by many companies which shows me that it's trustworthy. They are open-source tools that provide reliable mail transfer and mailbox management. Using both will cover all our bases, so there won't be any problems in the long run.

This will reduce expenses while still offering secure encrypted email and spam filtering from SpamAssassin. Since we will have some employees in the office and some at home, we need to make sure everyone can connect from various devices. This is the Linux strongsuit.

Hardware Requirements:

CPU:

- Quad-core 2.4 GHz or higher

RAM:

- Minimum 16 GB
- 32 to be safe

Storage:

- 1 TB SSD

Network:

- Gbps network interface card

Backup System:

- Cloud-based backup solution

These specifications ready us for steady flows of email traffic and mailbox storage. Since we have 40 employees, this should be perfect. It also has enough for potential new employees. If our company starts growing, we need to make sure that we are ready for it.

Web Server Implementation:

Operating System

- Linux (Ubuntu Server LTS)

Web Server Software

- Nginx

Homey Analytics will have a client dashboard and marketing website built with modern web tools. Nginx is known for high performance, low resource usage, and great connection upkeep. These 3 reasons make it ideal for client portals and remote access. It also supports reverse proxy configurations and SSL/TLS encryption. These are essential for secure client interactions.

Hardware Requirements

For hosting the website and client portal:

CPU:

- 6-core 3.0 GHz processor

RAM:

- 32 GB

Storage:

- 1 TB NVMe SSD

Network:

- 1 Gbps or higher NIC

It might be good idea to have a secondary server. This would allow for load balance and high availability, ensuring that our servers are up and running 24/7.

These requirements give us optimal performance for web traffic and secure client logins.

Conclusion

Homey analytics' industry is data analytics and consulting. This requires secure communication, strong data protection, and high uptime. Since Homey Analytics works

with sensitive retail data, security and reliability are the most important factors. For this reason, I chose Linux-based servers with open-source and well-supported software. Cost efficiency was also important due to the business being on the smaller side.

If the number of employees doubled to 80 over the next year, the company would need to increase storage capacity and upgrade RAM. For the email server, clustering or a cloud-based email model would work well. For the web server, load balancing and multiple web nodes would be needed to handle increased client traffic.